

MACON COUNTY AP, NC, USA

WMO: 720259

Lat: 35.223N

Lon: 83.419W

Elev: 616

StdP: 94.14

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63844

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-7.7	-19.8	0.7	-0.5	-16.5	1.0	-0.1	7.7	3.5	7.1	3.3	0.9	350	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	31.9	20.8	30.8	20.5	29.7	20.5	23.1	27.8	22.5	27.2	21.9	26.5	2.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.9	24.8	21.2	17.1	24.2	20.6	16.5	23.8	71.8	28.2	69.4	26.9	67.1	26.9	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.2	4.5	-13.5	33.6	3.0	1.9	-15.7	35.0	-17.4	36.0	-19.1	37.1	-21.3	38.5		
(5)			-14.0	24.0	2.9	1.0	-16.1	24.7	-17.8	25.3	-19.4	25.9	-21.6	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	3.5	5.3	9.0	13.5	17.7	22.0	23.4	22.9	20.2	14.3	8.4	5.4
	DBStd	8.08	5.35	4.97	4.58	3.84	3.44	2.24	2.02	2.21	2.71	4.41	4.39	4.80
	HDD10.0	697	212	146	74	13	1	0	0	0	0	12	81	158
	HDD18.3	2173	461	366	289	147	52	3	1	1	14	139	299	402
	CDD10.0	2092	9	13	43	118	241	358	415	399	306	144	31	15
	CDD18.3	529	0	0	0	2	33	111	157	141	71	13	1	0
	CDH23.3	4181	0	1	19	131	400	929	1115	971	508	102	5	0
	CDH26.7	1238	0	0	1	15	84	306	381	302	135	14	0	0
Wind	WSAvg	1.0	1.3	1.5	1.5	1.5	1.0	0.7	0.5	0.5	0.5	0.8	1.0	1.0
Precipitation	PrecAvg	1780	177	166	177	135	116	157	150	143	146	107	134	176
	PrecMax	2192	343	309	306	236	256	307	256	282	468	265	305	289
	PrecMin	1349	70	85	53	55	22	40	70	27	8	4	52	105
	PrecStd	279	67	57	64	49	55	64	53	69	111	64	59	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	22.1	25.8	28.2	30.9	32.9	32.7	32.7	31.3	28.4	23.9	20.6
		MCWB	12.3	15.1	13.8	15.1	17.6	19.6	21.3	21.5	19.9	19.2	14.3	14.0
	2%	DB	17.2	18.9	22.9	26.7	28.9	31.3	31.3	31.3	29.9	26.0	21.2	17.7
		MCWB	12.0	12.8	12.9	14.6	18.1	19.8	21.4	21.3	19.7	16.9	13.1	13.4
	5%	DB	15.0	17.1	21.0	24.9	27.4	30.0	30.2	29.9	28.0	24.0	19.0	16.0
		MCWB	11.4	10.9	12.8	14.3	17.3	20.0	21.1	21.2	19.0	15.9	12.7	12.2
	10%	DB	12.5	14.6	18.2	22.8	25.9	28.1	28.9	28.2	26.6	22.3	17.1	13.8
		MCWB	9.5	9.4	11.0	13.7	17.2	19.3	20.9	20.4	18.7	15.6	11.6	10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	16.4	16.6	19.1	22.2	22.7	24.0	24.0	22.3	21.3	17.9	16.8
		MCDB	16.6	19.5	22.2	23.3	27.4	27.7	29.1	29.5	27.2	25.2	20.7	18.7
	2%	WB	13.4	14.7	15.0	17.4	20.6	21.9	23.2	23.1	21.2	19.7	16.4	15.1
		MCDB	16.1	18.0	20.4	21.9	25.0	27.4	28.0	27.9	25.9	23.2	19.3	17.1
	5%	WB	12.0	12.7	13.9	16.4	19.6	21.3	22.5	22.6	20.5	18.6	14.5	13.1
		MCDB	14.0	15.3	18.7	21.9	24.1	26.8	27.2	27.0	25.1	22.1	17.0	15.0
	10%	WB	10.2	10.5	12.6	15.3	18.7	20.7	21.9	21.9	19.9	17.1	12.8	11.5
		MCDB	12.0	13.2	16.5	20.3	23.1	25.9	26.6	26.0	24.2	20.3	15.4	13.2
Mean Daily Temperature Range	5% DB	MDBR	12.5	13.0	14.1	15.3	13.9	12.7	11.5	11.6	12.4	14.2	15.1	12.3
		MCDBR	14.6	16.5	18.0	19.5	16.8	15.0	13.0	12.8	14.4	15.6	18.3	13.9
		MCWBR	9.7	9.9	9.2	8.9	7.2	5.5	4.8	4.7	5.8	7.5	10.3	9.2
	5% WB	MCDBR	10.9	11.9	14.8	14.4	12.5	12.3	11.1	10.5	12.1	11.9	11.3	9.9
		MCWBR	8.6	8.7	8.8	7.8	6.2	5.1	4.6	4.3	5.4	6.3	7.7	7.7
Clear-Sky Solar Irradiance	taub	0.300	0.310	0.341	0.377	0.433	0.459	0.497	0.500	0.395	0.362	0.322	0.299	
	taud	2.533	2.501	2.437	2.337	2.201	2.155	2.075	2.050	2.329	2.415	2.492	2.561	
	Ebn at Noon	911	939	930	907	857	831	796	785	862	864	873	890	
	Edh at Noon	80	92	107	125	145	152	163	165	118	100	83	74	
All-Sky Solar Radiation	RadAvg	2.51	3.08	4.19	5.25	5.78	6.07	5.77	5.37	4.66	3.85	2.85	2.16	
	RadStd	0.17	0.36	0.31	0.38	0.40	0.44	0.41	0.28	0.41	0.41	0.29	0.21	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	+145	+96

Nomenclature: See separate page